



AWS Database Migration Service (DMS)

Lecture Slides

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Good luck on your exam!

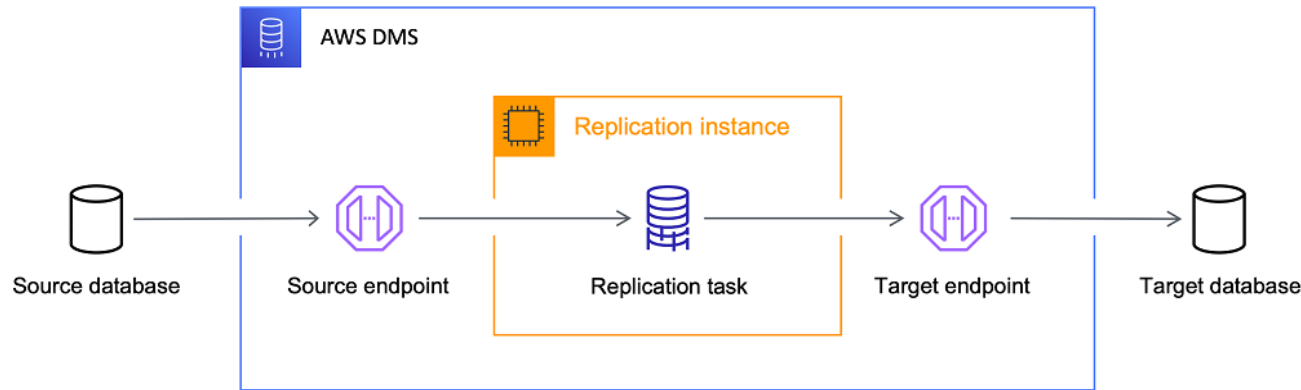
Database Migration Service (DMS)



AWS Database Migration Service (DMS) allows you to quickly and securely migrate one database to another. DMS can be used to migrate your on-premise database to AWS.

Possible Sources:

- Oracle Database
- Microsoft SQL
- MySQL
- MariaDB
- PostgreSQL
- MongoDB
- SAP ASE
- IMDB Db2
- Azure SQL Database
- Amazon RDS
- Amazon S3 (database dumps)
- Amazon Aurora
- Amazon DocumentDB



AWS Schema Conversion is used in many cases to automatically convert a source database schema to a target database schema.

For data warehouses you can use the desktop app **AWS Schema Conversion Tool**

Each migration path requires a bit of research since not all combinations of sources and targets are possible.

Possible Targets:

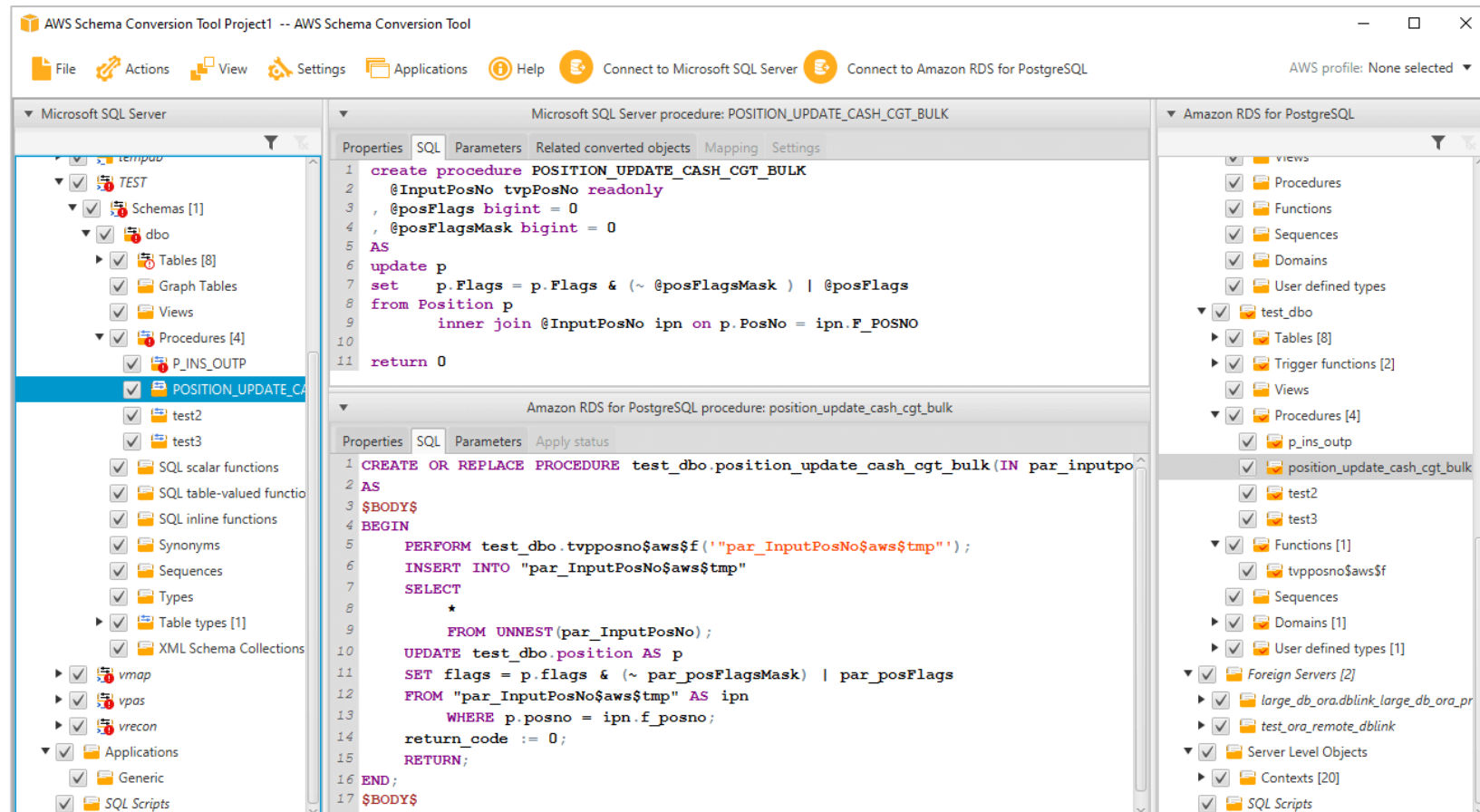
- Oracle Database
- Microsoft SQL
- MySQL
- MariaDB
- PostgreSQL
- Redis
- SAP ASE
- Amazon Redshift
- Amazon RDS
- Amazon DynamoDB
- Amazon S3
- Amazon Aurora
- Amazon OpenSearch Service
- Amazon ElastiCache for Redis
- Amazon DocumentDB
- Amazon Neptune
- Apache Kafka

Database Migration Service (DMS)

- **Migration Methods**
 - **Homogenous data migration**
 - Migrate data with native database tools. Eg. pg_dump, pg_restore
 - You create a migration project in DMS and it will perform the migration using a serverless compute
 - Uses a pay as you go model
 - **Instance Replication**
 - provision an instance with chosen instance type to perform the replications between databases
 - **Serverless Replication** (DMS Serverless)
 - a serverless offering where you pay as you go, with some limitations:
 - Does not have public Ips
 - Must use VPC endpoints to access specific AWS services. Eg. S3, Kinesis, DynamoDB, OpenSearch
 - Limited selection of possible sources and targets
 - Doesn't support views with selection and transformation rules.

AWS Schema Conversion Tool

AWS Schema Conversion Tool (AWS SCT) is a **stand-alone desktop app** convert your database schema to another database engine.



AWS SCT can be installed on:

- Windows
- Linux (Fedora or Ubuntu)
- ~~Mac (Not available for Mac)~~